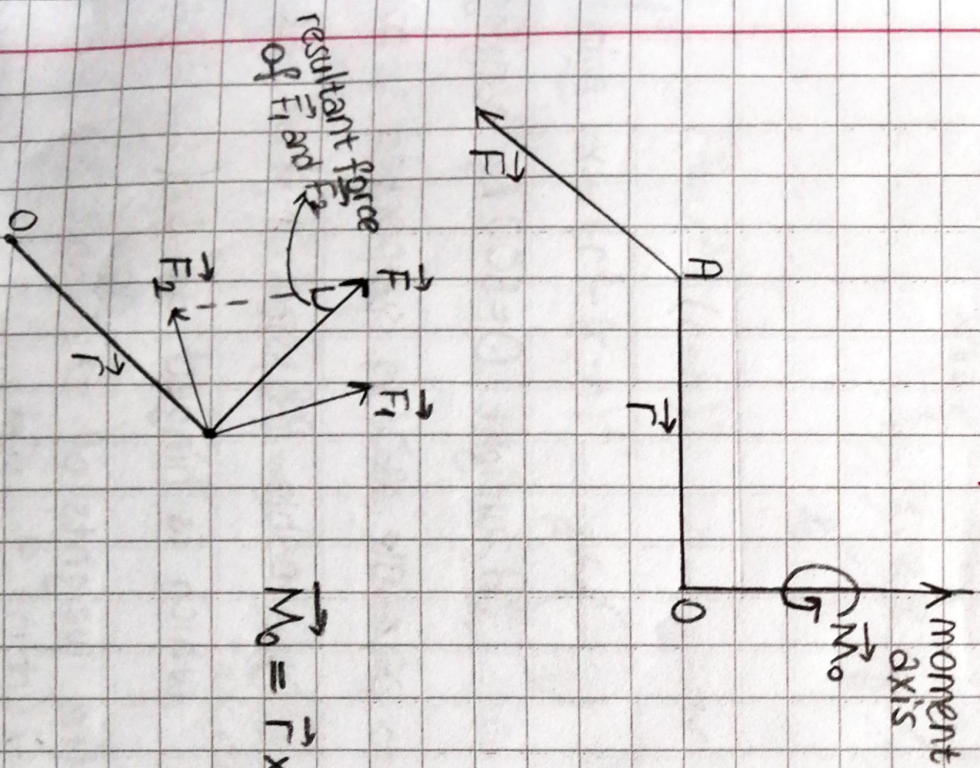


Moment of a Force- Vector Formulation



$$\vec{M}_O = \vec{r} \times \vec{F} =$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$$\vec{M}_O = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 = \vec{r} \times \vec{F} \quad (\text{Theorem of Varignon})$$